Jiani (Elva) He

Data Science Consultant · Independent Researcher
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RESEARCH INTERESTS

Demand-forecast stability and planning value; causal decision support under partial information; trustworthy deployment of AI systems; AI-assisted reverse logistics.

Related areas on the U.S. Critical and Emerging Technologies List (2026): AI and Autonomy (research), and Semiconductors and Microelectronics (through data center hardware supply chains).

PROFESSIONAL EXPERIENCE

Accenture
Jun 2024 - Present

Data Science Consultant — Seattle, Washington

Forward and reverse logistics in a technology client's data center supply chain, connecting operational analysis with planning and execution.

- Lead planning analyses, manage client relationships, and present recommendations to senior leaders.
- Analyze reverse-logistics cycle time and on-time delivery in SQL to identify bottlenecks; guide reuse, resale, recycling, or disposal decisions based on asset value and lifecycle stage.
- Analyze stock targets, available inventory, and demand; use an optimization engine to rebalance inventory across sites, address shortages, and maintain adequate supply.
- Plan quality-assurance component quantities using demand and bill-of-materials data; define service metrics and support automated inventory adjustments.
- Standardize prompts and analytical queries for enterprise LLM assistants; develop reusable guides and train analysts in AI-assisted analysis.

Accenture
Oct 2021 - May 2024

Data Science Senior Analyst — Seattle, Washington

- Built SQL dashboards and shortage alerts to compare inventory with demand and inform transfers between sites.
- Tracked reverse-logistics cycle times and on-time delivery; flagged bottlenecks and supported asset reuse, disposal, and recovery scheduling.
- Investigated inventory imbalances and abnormal cycle times using SQL and statistical analysis; automated reporting and purchase-order workflows.
- Across both Accenture roles, mentored five junior analysts in SQL, business problem solving, and client communication; led onboarding and team training.

BMW Group
Feb - Aug 2021

Data Scientist Intern · MIT Capstone, Dynamic Vehicle Pricing — Cambridge, Massachusetts

- Modeled vehicle demand and cross-price elasticity in Python to recommend discounts by customer segment and sales channel; built Tableau pricing dashboards.

EDUCATION

Massachusetts Institute of Technology
Aug 2021

MS, Business Analytics · Operations Research Center | GPA 4.7 / 5.0 — Cambridge, Massachusetts

Classification trees for 40,000 news articles (Python/Julia) and robust bike-allocation models (Julia). Analytics Lab: an NLP admissions Q&A pipeline for AdmitHub with response evaluation and human review.

Purdue University May 2020
BS, Industrial Engineering · Honors & Distinction | GPA 3.9 / 4.0 West Lafayette, Indiana
LSTM forecasting models (Keras) and reinforcement learning for sensor placement (Python/R).

RESEARCH & PUBLICATIONS

Lead-author papers

1. **Jiani He**, Dingyan Shang, Yihua Xu, Shiqi Huang, Yan Lyu, Jize Li, Shangjing Tang. 2026. "Semantic Signal-Assisted Inspection and Recovery Allocation in Reverse Logistics." *4th International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings 2026)*. arXiv:2609.02116. **Accepted · AIBThings 2026; proceedings forthcoming**

2. Shiqi Huang*, **Jiani He***, Dingyan Shang, Yihua Xu, Jize Li, Yan Lyu, Lashimi Muraleedharan Nair (*equal contribution). 2026. "DACRI: Decision-Aware Causal Intervention Ranking for Critical Supply Chains." *International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME 2026) · 15–17 October 2026 · Bali, Indonesia*. arXiv:2608.11154. **Accepted · ICECCME 2026; proceedings forthcoming**

Accepted conference papers

3. Yihua Xu, Yan Lyu, Jize Li, **Jiani He**, You Lin, Yidan Hu, Rubing Qiang. 2026. "When Does an Added Planning Capability Create Operational Value? An Observable Evidence Standard." *3rd World Congress on Industrial Engineering and Operations Management · Irvine, California*. arXiv:2610.27305. **Accepted · IEOM Irvine 2026; proceedings forthcoming**

4. Yan Lyu, Yihua Xu, **Jiani He**, Jize Li, Rubing Qiang, Yidan Hu. 2026. "From Weak Signals to Supply Chain Reliability Risk: A Feedback-Driven Risk Accumulation Framework for Late-Delivery Prediction, Early Warning, and Risk Prioritization." *3rd World Congress on Industrial Engineering and Operations Management · Irvine, California*. SSRN preprint submitted · posting pending. **Accepted · IEOM Irvine 2026; proceedings forthcoming**

5. Jize Li, **Jiani He**, Dishu Yang, Dingyan Shang, Jingjing Liu, Shiqi Huang. 2026. "Accuracy-Preserving Stability Regularization for Large-Scale Retail Demand Forecasting." *ICEME 2026*. arXiv:2607.13331. **Accepted · ICEME 2026; proceedings forthcoming**

6. Yihua Xu, **Jiani He**, Ishita Chirag Talati, Yitian Qian, Youting Wang, Zhenyu Xu, Dingyan Shang. 2026. "Policy-Bounded, Provenance-Aware Deployment Assurance for LLM Request Routing." *ICOAI 2026 · public preprint on SSRN*. SSRN 7554440. **Accepted · ICOAI 2026; proceedings forthcoming**

SERVICE & MENTORING

- Peer reviewer for 10+ journal manuscripts.
- Mentored five junior analysts across Accenture roles; led onboarding and team training, including AI-assisted analysis.

SKILLS

Tools: SQL, Python, R, Julia, Keras, Tableau, AWS

Modeling: Machine learning, Forecasting, Regression, Classification, Clustering, Robust optimization

Analytics & BI: Statistical analysis, Root-cause analysis, Data visualization, Dashboards, KPI design

AI: Generative AI, NLP, LLM workflows, Prompt & query design

Consulting: Problem framing, Client communication, Stakeholder coordination, Project delivery, Mentoring